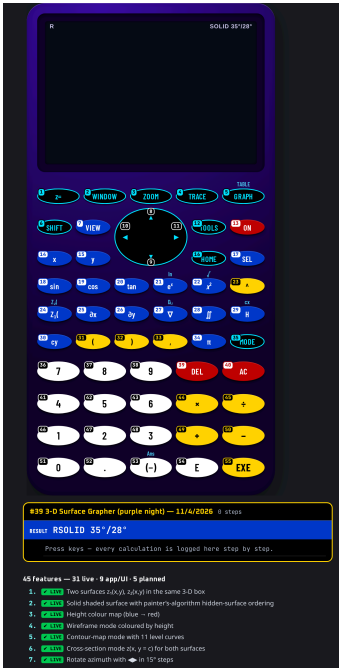
U-45 CALCULATORS

User Manual

Calculator #39 3-D Surface Grapher (purple night)

Graphing – colour

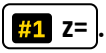


Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#39 — 3-D Surface Grapher (purple night)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example  . Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

3-D surface graphing z = f(x,y) with rotation, contours and cross-sections.

Chapters

1. Getting started
2. Features
3. Feature guide (2b)
4. Numbered key map
5. Key-by-key reference
6. Worked examples
7. Tricks & hacks
8. Keyboard shortcuts
9. Common mistakes
10. Practice drills
11. Appendix A — Drill answers
12. Appendix B — Function finder (A–Z)

13. Appendix C — Glossary

14. Appendix D — Model notes & limits

15. Appendix E — Quick-reference card

1 · Getting started

Open U-45 CALCULATORS, choose **#39** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#36, #37, #38, #41, #42, #43, #46, #47, #48, #51, #52, #53, #54
op	Operators & execute	#23, #31, #32, #33, #44, #45, #49, #50, #55
fn	Functions	#7, #14, #15, #17, #18, #19, #20, #21, #22, #24, #25, #26, #27, #28, #29, #30, #34
mem	Memory, variables & registers	—
clr	Clear, delete & power	#13, #39, #40

mod	**Shift, modes, menus & navigation**	**#1, #2, #3, #4, #5, #6, #8, #9, #10, #11, #12, #16, #35**

2 · Features

45 features: 31 live on the keys, 9 app/UI features, 5 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE Two surfaces $z_1(x,y)$, $z_2(x,y)$ in the same 3-D box
2. LIVE Solid shaded surface with painter's-algorithm hidden-surface ordering
3. LIVE Height colour map (blue → red)
4. LIVE Wireframe mode coloured by height
5. LIVE Contour-map mode with 11 level curves
6. LIVE Cross-section mode z(x, y = c) for both surfaces
7. LIVE Rotate azimuth with ◀▶ in 15° steps
8. LIVE Tilt elevation with ▲ ▼ in 10° steps
9. LIVE Zoom closer/farther
10. LIVE Top view and side view presets
11. LIVE Automatic z-range option
12. LIVE Adjustable grid resolution 6–60
13. LIVE Bounding box with axis labels
14. LIVE 3-D trace cursor moving over the x-y grid
15. LIVE Trace read-out of z and both partial derivatives
16. LIVE Tool: value z(x,y)
17. LIVE Tool: partial derivatives and gradient magnitude
18. LIVE Tool: tangent-plane equation
19. LIVE Tool: directional derivative along (ux,uy)
20. LIVE Tool: critical point by Newton's method with second-derivative test

21. LIVE Tool: double integral over the window (nested Simpson)
22. LIVE Tool: volume between two surfaces
23. LIVE Tool: grid maximum and minimum with markers
24. LIVE VIEW key cycles SOLID → WIRE → CONTOUR → SECTION
25. LIVE Table of sections z(x,0), z(x,1), z(0,x)
26. LIVE Home Z1(), Z2(), dzdx(), dzdy(), gnorm()
27. LIVE Home directional derivative dirder(x,y,a,b)
28. LIVE Home Hessian determinant hessd(x,y)
29. LIVE Home double integral dint(a,b,c,d)
30. LIVE Home critical-point coordinates crx(), cry()
31. LIVE Night-sky display distinct from the 2-D graphers
32. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
33. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
34. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
35. APP/UI Show / hide key numbers with one button (remembered between sessions)
36. APP/UI Copy the current display text to the clipboard
37. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
38. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
39. APP/UI Reference photo viewer and tool-tip name on every key

40. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
41. PLANNED **Graph image export (PNG)**
42. PLANNED **Shareable function list link**
43. PLANNED **Dark screen theme**
44. PLANNED **Speech read-out of values**
45. PLANNED **Exam mode**

2b · Feature guide

Calculator #39 is a 3-D surface graphing z = f(x,y) with rotation, contours and cross-sections. Each card below is one of its 31 working features, with the keys that reach it and, where one exists, a verified example.

F1 Two surfaces $z_1(x,y)$, $z_2(x,y)$ in the same 3-D box

Where: [#1 z=] Edit surfaces; [#52 .] Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F2 Solid shaded surface with painter's-algorithm hidden-surface ordering

Where: [#24 Z_1(] Height of surface 1; [#7 VIEW] Solid / wire / contour / section

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F3 Height colour map (blue → red)

Where: #24 $Z_1($ Height of surface 1

See it in action: Height of z = sin x cos y at (π/2, 0) — *Z1(π/2,0)* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F4 Wireframe mode coloured by height

Where: #6 SHIFT SHIFT / 2nd — the next key uses its upper (shift-coloured) function; #24 $Z_1($ Height of surface 1

See it in action: Height of z = sin x cos y at (π/2, 0) — *Z1(π/2,0)* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F5 Contour-map mode with 11 level curves

Where: #7 VIEW Solid / wire / contour / section

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F6 Cross-section mode z(x, y = c) for both surfaces

Where: #1 z= Edit surfaces; #7 VIEW Solid / wire / contour / section

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F7 Rotate azimuth with ◀▶ in 15° steps

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F8 Tilt elevation with ▲ ▼ in 10° steps

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F9 Zoom closer/farther

Where: #3 ZOOM ZOOM — zoom presets

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F10 Top view and side view presets

Where: #7 VIEW Solid / wire / contour / section; #3 ZOOM ZOOM — zoom presets

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F11 Automatic z-range option

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F12 Adjustable grid resolution 6–60

Where: #2 WINDOW x, y, z ranges and grid

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F13 Bounding box with axis labels

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F14 3-D trace cursor moving over the x-y grid

Where: #2 WINDOW x, y, z ranges and grid; #4 TRACE TRACE — move along a graph; #8 ▲ Cursor ▲ — up / previous history line / previous item

See it in action: Double integral of sin x cos y over [0,π]×[0,π/2] — *dint(0,π,0,π/2)* = 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F15 Trace read-out of z and both partial derivatives

Where: #4 TRACE TRACE — move along a graph; #25 ∂x Partial derivative in x; #26 ∂y Partial derivative in y

See it in action: Partial derivative ∂z/∂x at the origin — *dzdx(0,0)* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F16 Tool: value z(x,y)

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F17 Tool: partial derivatives and gradient magnitude

Where: #27 ∇ Gradient magnitude; #25 ∂x Partial derivative in x; #26 ∂y Partial derivative in y

See it in action: Partial derivative ∂z/∂x at the origin — *dzdx(0,0)* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F18 Tool: tangent-plane equation

Where: #20 tan Tangent (current angle unit)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F19 Tool: directional derivative along (ux,uy)

Where: #27 ∇ Gradient magnitude; #4 TRACE TRACE — move along a graph; #25 ∂x Partial derivative in x

See it in action: Partial derivative ∂z/∂x at the origin — *dzdx(0,0)* = 1; Directional derivative at the origin toward (1,1) — *dirder(0,0,1,1)* = 0.70710678

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F20 Tool: critical point by Newton's method with second-derivative test

Where: #29 H Hessian determinant; #30 cy Critical point y; #52 . Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

See it in action: Partial derivative ∂z/∂x at the origin — *dzdx(0,0)* = 1; Directional derivative at the origin toward (1,1) — *dirder(0,0,1,1)* = 0.70710678

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F21 Tool: double integral over the window (nested Simpson)

Where: #28 ∬ Double integral of z_1; #2 WINDOW x, y, z ranges and grid

See it in action: Double integral of sin x cos y over [0,π]×[0,π/2] — *dint(0,π,0,π/2)* = 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F22 Tool: volume between two surfaces

Where: #1 z= Edit surfaces; #33 , Comma / argument separator (between function arguments, list items or coordinates)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F23 Tool: grid maximum and minimum with markers

Where: #2 WINDOW x, y, z ranges and grid

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F24 VIEW key cycles SOLID → WIRE → CONTOUR → SECTION

Where: #7 VIEW Solid / wire / contour / section

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F25 Table of sections z(x,0), z(x,1), z(0,x)

Where: #5 GRAPH GRAPH — draw the graph

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F26 Home Z1(), Z2(), dzdx(), dzdy(), gnorm()

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

See it in action: Partial derivative ∂z/∂x at the origin — *dzdx(0,0)* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F27 Home directional derivative dirder(x,y,a,b)

Where: #27 ∇ Gradient magnitude; #25 ∂x Partial derivative in x; #26 ∂y Partial derivative in y

See it in action: Partial derivative ∂z/∂x at the origin — *dzdx(0,0)* = 1; Directional derivative at the origin toward (1,1) — *dirder(0,0,1,1)* = 0.70710678

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F28 Home Hessian determinant hessd(x,y)

Where: #29 H Hessian determinant

See it in action: Hessian determinant at the peak (π/2, 0) — *hessd(π/2,0)* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F29 Home double integral dint(a,b,c,d)

Where: #28 ∬ Double integral of z_1

See it in action: Double integral of sin x cos y over [0,π]×[0,π/2] — *dint(0,π,0,π/2)* = 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F30 Home critical-point coordinates crx(), cry()

Where: #29 H Hessian determinant; #30 cy Critical point y; #33 , Comma / argument separator (between function arguments, list items or coordinates)

See it in action: Critical point near (1.4, 0.2): x-coordinate (= π/2) — *crx(1.4,0.2)* = 1.57079633

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F31 Night-sky display distinct from the 2-D graphers

Where: #35 MODE **SETUP — angle unit, number format, display options**

Try it: Press Reset, use the key(s) above, and write the display here:

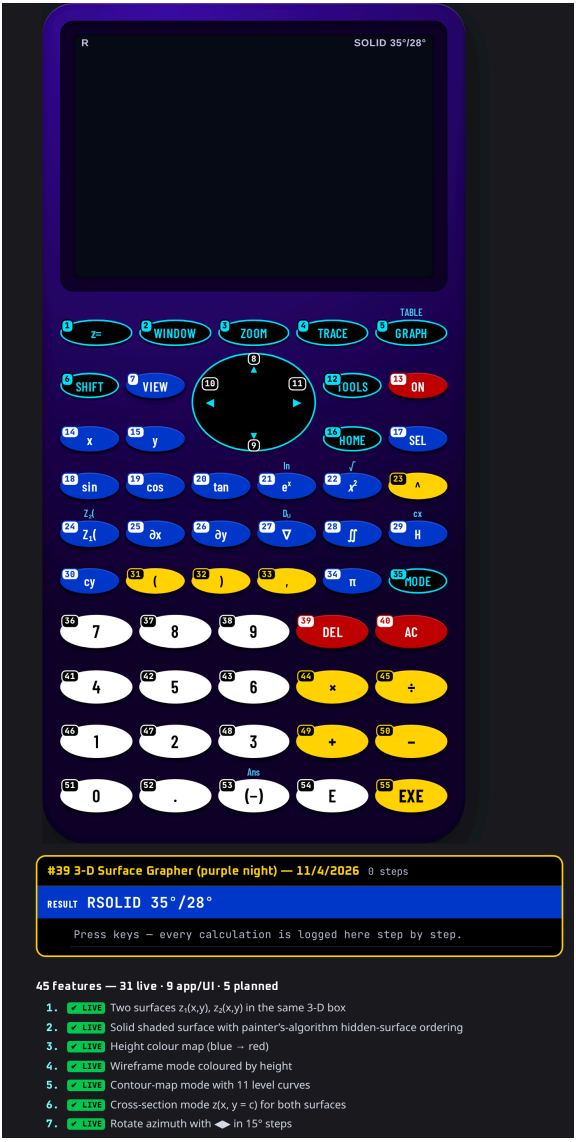
_____

Your notes:

_____

_____

3 · Numbered key map



Calculator #39 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	z=	·	·	·	mod
2	WINDOW	·	·	·	mod
3	ZOOM	·	·	·	mod
4	TRACE	·	·	·	mod
5	GRAPH	TABLE	·	·	mod
6	SHIFT	·	·	·	mod
7	VIEW	·	·	·	fn
8	▲	·	·	·	mod
9	▼	·	·	·	mod
10	◀	·	·	·	mod
11	▶	·	·	·	mod
12	TOOLS	·	·	·	mod
13	ON	·	·	·	clr
14	x	·	·	·	fn
15	y	·	·	·	fn
16	HOME	·	·	·	mod
17	SEL	·	·	·	fn
18	sin	·	·	·	fn
19	cos	·	·	·	fn
20	tan	·	·	·	fn
21	ex	ln	·	·	fn
22	x^2	√	·	·	fn
23	^	·	·	·	op

#	Face	Shift	Alpha	3rd	Group
24	Z_1(	Z_2(	·	·	fn
25	∂x	·	·	·	fn
26	∂y	·	·	·	fn
27	∇	D_u	·	·	fn
28	∬	·	·	·	fn
29	H	cx	·	·	fn
30	cy	·	·	·	fn
31	(	·	·	·	op
32	)	·	·	·	op
33	,	·	·	·	op
34	π	·	·	·	fn
35	MODE	·	·	·	mod
36	7	·	·	·	num
37	8	·	·	·	num
38	9	·	·	·	num
39	DEL	·	·	·	clr
40	AC	·	·	·	clr
41	4	·	·	·	num
42	5	·	·	·	num
43	6	·	·	·	num
44	×	·	·	·	op
45	÷	·	·	·	op
46	1	·	·	·	num
47	2	·	·	·	num
48	3	·	·	·	num
49	+	·	·	·	op
50	−	·	·	·	op
51	0	·	·	·	num

#	Face	Shift	Alpha	3rd	Group
52	.	·	·	·	num
53	(–)	Ans	·	·	num
54	E	·	·	·	num
55	EXE	·	·	·	op

4 · Key-by-key reference

Every key on calculator #39, in number order, with every function it carries in this app.

#1 z= `Shift, modes, menus & navigation`

Primary: Edit surfaces

#2 WINDOW `Shift, modes, menus & navigation`

Primary: x, y, z ranges and grid

#3 ZOOM `Shift, modes, menus & navigation`

Primary: ZOOM — zoom presets

#4 TRACE `Shift, modes, menus & navigation`

Primary: TRACE — move along a graph

#5 GRAPH `Shift, modes, menus & navigation`

Primary: GRAPH — draw the graph

shifted (2nd / f / SHIFT layer) (press #6 first): TABLE — TABLE — table of values

#6 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#7 VIEW Functions

Primary: Solid / wire / contour / section

#8 ▲ Shift, modes, menus & navigation

Primary: Cursor ▲ — up / previous history line / previous item

#9 ▼ Shift, modes, menus & navigation

Primary: Cursor ▼ — down / next history line / next item

#10 ◀ Shift, modes, menus & navigation

Primary: Cursor ◀ — move left in the line

#11 ▶ Shift, modes, menus & navigation

Primary: Cursor ▶ — move right in the line

#12 TOOLS Shift, modes, menus & navigation

Primary: TOOLS — tools menu

#13 ON Clear, delete & power

Primary: ON — power on

#14 x Functions

Primary: Variable x

#15 y Functions

Primary: Variable y

#16 HOME Shift, modes, menus & navigation

Primary: HOME — return to the main calculation screen

#17 SEL Functions

Primary: sel

#18 sin Functions

Primary: Sine (uses the current angle unit)

Keyboard: S

#19 cos Functions

Primary: Cosine (current angle unit)

Keyboard: C

#20 tan Functions

Primary: Tangent (current angle unit)

Keyboard: t

#21 ex Functions

Primary: e^x — exponential

shifted (2nd / f / SHIFT layer) (press #6 first): ln — Natural logarithm ln

#22 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #6 first): √ — Square root √

Keyboard: q

#23 ^ Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

Keyboard: ^

#24 Z_1(Functions

Primary: Height of surface 1

shifted (2nd / f / SHIFT layer) (press #6 first): Z_2(— Height of surface 2

#25 ∂x Functions

Primary: Partial derivative in x

#26 ∂y Functions

Primary: Partial derivative in y

#27 ∇ Functions

Primary: Gradient magnitude

shifted (2nd / f / SHIFT layer) (press #6 first): D_u — Directional derivative

#28 ∬ Functions

Primary: Double integral of z_1

#29 H Functions

Primary: Hessian determinant

shifted (2nd / f / SHIFT layer) (press #6 first): cx — Critical point x

#30 cy Functions

Primary: Critical point y

#31 (Operators & execute

Primary: Opening bracket (

Keyboard: (

#32) Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

Keyboard:)

#33 , Operators & execute

Primary: Comma / argument separator (between function arguments, list items or coordinates)

Keyboard: ,

#34 π Functions

Primary: π (3.14159265…) as an exact constant

Keyboard: p

#35 MODE Shift, modes, menus & navigation

Primary: SETUP — angle unit, number format, display options

#36 7 Digits & number entry

Primary: Digit 7

Keyboard: 7

#37 8 Digits & number entry

Primary: Digit 8

Keyboard: 8

#38 9 Digits & number entry

Primary: Digit 9

Keyboard: 9

#39 DEL Clear, delete & power

Primary: DEL — delete the character at the cursor

Keyboard: Backspace Delete

#40 AC Clear, delete & power

Primary: AC — clear all of the current calculation

Keyboard: Escape

#41 4 Digits & number entry

Primary: Digit 4

Keyboard: 4

#42 5 Digits & number entry

Primary: Digit 5

Keyboard: 5

#43 6 Digits & number entry

Primary: Digit 6

Keyboard: 6

#44 × Operators & execute

Primary: Multiplication ×

Keyboard: * x

#45 ÷ Operators & execute

Primary: Division ÷

Keyboard: /

#46 1 Digits & number entry

Primary: Digit 1

Keyboard: 1

#47 2 Digits & number entry

Primary: Digit 2

Keyboard: 2

#48 3 Digits & number entry

Primary: Digit 3

Keyboard: 3

#49 **+** `Operators & execute`

Primary: Addition +

Keyboard: `+`

#50 **–** `Operators & execute`

Primary: Subtraction –

Keyboard: `-`

#51 **0** `Digits & number entry`

Primary: Digit 0

Keyboard: `0`

#52 **.** `Digits & number entry`

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

Keyboard: `.`

#53 **(–)** `Digits & number entry`

Primary: Negative sign (–) — makes the next number negative. Different from the subtraction key

shifted (2nd / f / SHIFT layer) (press #6 first): Ans — Ans — recall the last answer into the current expression

#54 E `Digits & number entry`

Primary: ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23

Keyboard: `e`

#55 EXE `Operators & execute`

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Keyboard: `Enter` `=`

5 · Worked examples

6 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Height of z = sin x cos y at (π/2, 0)

Calculate **Z1(π/2,0)**. Press **Reset** first.

#24 $Z_1($ → #34 π → #45 ÷ → #47 2 → #33 , → #51 0 → #32) → #55 EXE

Display shows `1` — expected 1. ✔ verified on the app.

Example 5.2 — Partial derivative ∂z/∂x at the origin

Calculate **dzdx(0,0)**. Press **Reset** first.

#25 ∂x → #51 0 → #33 , → #51 0 → #32) → #55 EXE

Display shows `1` — expected 1. ✔ verified on the app.

Example 5.3 — Double integral of sin x cos y over [0,π]×[0,π/2]

Calculate **dint(0,π,0,π/2)**. Press **Reset** first.

#28 ∬ → #51 0 → #33 , → #34 π → #33 , → #51 0 → #33 , → #34 π → #45 ÷ → #47 2 → #32) → #55 EXE

Display shows `2.000000028` — expected 2. ✔ verified on the app.

Example 5.4 — Hessian determinant at the peak (π/2, 0)

Calculate **hessd(π/2,0)**. Press **Reset** first.

#29 H → #34 π → #45 ÷ → #47 2 → #33 , → #51 0 → #32) → #55 EXE

Display shows `0.9999998333` — expected 1. ✔ verified on the app.

Example 5.5 — Directional derivative at the origin toward (1,1)

Calculate **dirder(0,0,1,1)**. Press **Reset** first.

#6 SHIFT → #27 D_u → #51 0 → #33 , → #51 0 → #33 , → #46 1 → #33 , → #46 1 → #32) → #55 EXE

Display shows `0.7071067812` — expected 0.70710678. ✔ verified on the app.

Example 5.6 — Critical point near (1.4, 0.2): x-coordinate (= π/2)

Calculate **crx(1.4,0.2)**. Press **Reset** first.

#6 SHIFT → #29 cx → #46 1 → #52 . → #41 4 → #33 , → #51 0 → #52 . → #47 2 → #32) → #55 EXE

Display shows `1.570796327` — expected 1.57079633. ✔ verified on the app.

6 · Tricks & hacks

Hack 1: History reuse

Scroll up in the home screen history and press ENTER/EXE to copy an old entry or answer.

Hack 2: Ans chaining

Start an entry with an operator to continue from the previous answer.

Hack 3: Store variables

Use STO▶ / → to store values into letters A–Z, then use them in expressions.

Hack 4: Table of values

Use the TABLE view to scan where a function changes sign before solving.

Hack 5: Window first

If a graph looks empty, adjust the WINDOW / zoom standard.

Hack 6: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 7: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = ...

Hack 8: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 9: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 10: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 11: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 12: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 13: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 14: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 15: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 16: Hidden functions on this model

7 extra functions sit on shifted layers. Press the layer key once, then the key: for example #6 then #5 → TABLE; #6 then #21 → ln; #6 then #22 → √; #6 then #24 → Z_2(.

Hack 17: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left→right, top→bottom from #1 to #55.

Hack 18: Type faster on a computer

28 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 19: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 20: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 21: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
s	#18	sin	Sine (uses the current angle unit)
c	#19	cos	Cosine (current angle unit)
t	#20	tan	Tangent (current angle unit)
q	#22	x^2	Square x^2
^	#23	^	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
(	#31	(	Opening bracket (
)	#32	)	Closing bracket). Closing brackets at the end of a line may be omitted
,	#33	,	Comma / argument separator (between function arguments, list items or coordinates)
p	#34	π	π (3.14159265…) as an exact constant
7	#36	7	Digit 7
8	#37	8	Digit 8
9	#38	9	Digit 9
Backspace Delete	#39	DEL	DEL — delete the character at the cursor
Escape	#40	AC	AC — clear all of the current calculation
4	#41	4	Digit 4
5	#42	5	Digit 5
6	#43	6	Digit 6
* x	#44	×	Multiplication ×
/	#45	÷	Division ÷
1	#46	1	Digit 1

2	#47	2	Digit 2
3	#48	3	Digit 3
+	#49	+	Addition +
-	#50	−	Subtraction −
0	#51	0	Digit 0
.	#52	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
e	#54	E	×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23
Enter =	#55	EXE	Execute / = — evaluate the line and show the result (EXE / ENTER / =)

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: (–) vs –

Use the negative key for negative numbers.

Mistake 2: Radian mode

Graphs of sin x in degrees look flat over −10..10; switch to radians for standard trig windows.

Mistake 3: Implicit multiplication

2/3x may be read as 2/(3x) depending on the model; use brackets.

Mistake 4: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 5: Wrong angle unit

If sin 30 gives −0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 6: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 7: Division precedence

1 ÷ 2π may be read as 1 ÷ (2π) on textbook models. Type 1 ÷ (2 × π) explicitly to be sure.

Mistake 8: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 9: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 10: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 11: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

Mistake 12: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

9 · Practice drills

Set A — Calculation drills (6)

Work each one on calculator #39, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. **Z1(π/2,0) =** _____
2. **dzdx(0,0) =** _____
3. **dint(0,π,0,π/2) =** _____
4. **hessd(π/2,0) =** _____
5. **dirder(0,0,1,1) =** _____
6. **crx(1.4,0.2) =** _____

Set B — Key-finding drills (52)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. **Which key is "z=" (Edit surfaces)?** _____
2. **Which key is "WINDOW" (x, y, z ranges and grid)?** _____
3. **Which key is "ZOOM" (ZOOM)?** _____
4. **Which key is "TRACE" (TRACE)?** _____
5. **Which key is "GRAPH" (GRAPH)?** _____
6. **Which key is "SHIFT" (SHIFT / 2nd)?** _____
7. **Which key is "VIEW" (Solid / wire / contour / section)?** _____
8. **Which key is " ▲ " (Cursor ▲)?** _____

9. Which key is “▼” (Cursor ▼)? _____

10. Which key is “◀” (Cursor ◀)? _____

11. Which key is “▶” (Cursor ▶)? _____

12. Which key is “TOOLS” (TOOLS)? _____

13. Which key is “ON” (ON)? _____

14. Which key is “x” (Variable x)? _____

15. Which key is “y” (Variable y)? _____

16. Which key is “HOME” (HOME)? _____

17. Which key is “SEL” (sel)? _____

18. Which key is “sin” (Sine)? _____

19. Which key is “cos” (Cosine)? _____

20. Which key is “tan” (Tangent)? _____

21. Which key is “ex” (e^x)? _____

22. Which key is “x^2” (Square x^2)? _____

23. Which key is “^” (Power x^y / ^)? _____

24. Which key is “Z_1(” (Height of surface 1)? _____

25. Which key is “∂x” (Partial derivative in x)? _____

26. Which key is “∂y” (Partial derivative in y)? _____

27. Which key is “∇” (Gradient magnitude)? _____

28. Which key is “∬” (Double integral of z_1)? _____

29. Which key is “H” (Hessian determinant)? _____

30. Which key is “cy” (Critical point y)? _____

31. Which key is “(” (Opening bracket)? _____

32. Which key is “)” (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

33. Which key is “,” (Comma / argument separator)? _____

34. Which key is “π” (π)? _____

35. Which key is “MODE” (SETUP)? _____

36. Which key is “DEL” (DEL)? _____

37. Which key is “AC” (AC)? _____

38. Which key is “×” (Multiplication ×)? _____

39. Which key is “÷” (Division ÷)? _____

40. Which key is “+” (Addition +)? _____

41. Which key is “−” (Subtraction −)? _____

42. Which key is “.” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

43. Which key is “(−)” (Negative sign)? _____

44. Which key is “E” (×10ˣ / EXP / EE)? _____

45. Which key is “EXE” (Execute / =)? _____

46. How do you reach “TABLE”? _____

47. How do you reach “ln”? _____

48. How do you reach "√"? _____

49. How do you reach "Z_2("? _____

50. How do you reach "D_U"? _____

51. How do you reach "cx"? _____

52. How do you reach "Ans"? _____

Appendix A — Drill answers

Set A

1. **1** (display `1`)
   `#24` $Z_1($ → `#34` π → `#45` ÷ → `#47` 2 → `#33` , → `#51` 0 → `#32`) → `#55` EXE

2. **1** (display `1`)
   `#25` ∂x → `#51` 0 → `#33` , → `#51` 0 → `#32`) → `#55` EXE

3. **2** (display `2.000000028`)
   `#28` ∬ → `#51` 0 → `#33` , → `#34` π → `#33` , → `#51` 0 → `#33` , → `#34` π → `#45` ÷ → `#47` 2 → `#32`) → `#55` EXE

4. **1** (display `0.9999998333`)
   `#29` H → `#34` π → `#45` ÷ → `#47` 2 → `#33` , → `#51` 0 → `#32`) → `#55` EXE

5. **0.70710678** (display `0.7071067812`)
   `#6` SHIFT → `#27` D_u → `#51` 0 → `#33` , → `#51` 0 → `#33` , → `#46` 1 → `#33` , → `#46` 1 → `#32`) → `#55` EXE

6. **1.57079633** (display `1.570796327`)
   `#6` SHIFT → `#29` cx → `#46` 1 → `#52` . → `#41` 4 → `#33` , → `#51` 0 → `#52` . → `#47` 2 → `#32`) → `#55` EXE

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #31
32. #32
33. #33
34. #34
35. #35
36. #39
37. #40
38. #44
39. #45
40. #49
41. #50
42. #52
43. #53
44. #54
45. #55
46. #6 SHIFT then #5 GRAPH
47. #6 SHIFT then #21 ex
48. #6 SHIFT then #22 x^2
49. #6 SHIFT then #24 Z_1(
50. #6 SHIFT then #27 ∇
51. #6 SHIFT then #29 H
52. #6 SHIFT then #53 (-)

Appendix B — Function finder (A–Z)

Function	Label	Keys
×10^x / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^x 23	E	#54
AC — clear all of the current calculation	AC	#40
Addition +	+	#49
Ans — recall the last answer into the current expression	Ans	#6 → #53
Closing bracket). Closing brackets at the end of a line may be omitted	)	#32
Comma / argument separator (between function arguments, list items or coordinates)	,	#33
Cosine (current angle unit)	cos	#19
Critical point x	cx	#6 → #29
Critical point y	cy	#30
Cursor ▲ — up / previous history line / previous item	▲	#8
Cursor ▶ — move right in the line	▶	#11
Cursor ▼ — down / next history line / next item	▼	#9
Cursor ◀ — move left in the line	◀	#10
Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored	.	#52
DEL — delete the character at the cursor	DEL	#39
Digit 0	0	#51
Digit 1	1	#46
Digit 2	2	#47
Digit 3	3	#48
Digit 4	4	#41
Digit 5	5	#42
Digit 6	6	#43

Digit 7	7	#36
Digit 8	8	#37
Digit 9	9	#38
Directional derivative	D_u	#6 → #27
Division ÷	÷	#45
Double integral of z_1	∬	#28
Edit surfaces	z=	#1
e^x — exponential	ex	#21
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	EXE	#55
Gradient magnitude	∇	#27
GRAPH — draw the graph	GRAPH	#5
Height of surface 1	Z_1(	#24
Height of surface 2	Z_2(	#6 → #24
Hessian determinant	H	#29
HOME — return to the main calculation screen	HOME	#16
Multiplication ×	×	#44
Natural logarithm ln	ln	#6 → #21
Negative sign (−) — makes the next number negative. Different from the subtraction key	(−)	#53
ON — power on	ON	#13
Opening bracket (	(	#31
Partial derivative in x	∂x	#25
Partial derivative in y	∂y	#26
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	^	#23
sel	SEL	#17
SETUP — angle unit, number format, display options	MODE	#35
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#6

Sine (uses the current angle unit)	sin	#18
Solid / wire / contour / section	VIEW	#7
Square root √	√	#6 → #22
Square x^2	x^2	#22
Subtraction –	–	#50
TABLE — table of values	TABLE	#6 → #5
Tangent (current angle unit)	tan	#20
TOOLS — tools menu	TOOLS	#12
TRACE — move along a graph	TRACE	#4
Variable x	x	#14
Variable y	y	#15
x, y, z ranges and grid	WINDOW	#2
ZOOM — zoom presets	ZOOM	#3
π (3.14159265…) as an exact constant	π	#34

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

3-D surface grapher. Arrow keys rotate the view; TRACE moves a cursor over the x-y grid.

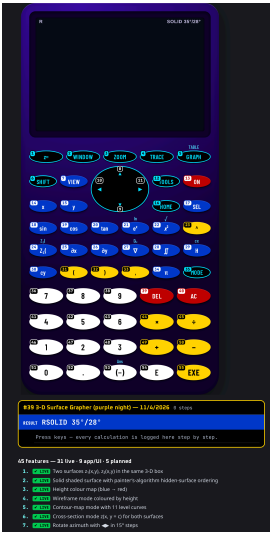
Double integrals use nested Simpson rules (80 × 80 panels); critical points use Newton's method on the gradient with a numerical Hessian.

Planned (not yet working)

- **Graph image export (PNG)**
- **Shareable function list link**
- **Dark screen theme**
- **Speech read-out of values**
- **Exam mode**

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 z=	#2 WINDOW	#3 ZOOM
#4 TRACE	#5 GRAPH ⇧ TABLE	#6 SHIFT
#7 VIEW	#8 ▲	#9 ▼
#10 ◀	#11 ▶	#12 TOOLS
#13 ON	#14 x	#15 y
#16 HOME	#17 SEL	#18 sin
#19 cos	#20 tan	#21 ex ⇧ ln
#22 x^2 ⇧ √	#23 ^	#24 Z_1(⇧ Z_2(
#25 ∂x	#26 ∂y	#27 ∇ ⇧ D_u
#28 ∬	#29 H ⇧ cx	#30 cy
#31 (	#32)	#33 ,
#34 π	#35 MODE	#36 7
#37 8	#38 9	#39 DEL
#40 AC	#41 4	#42 5
#43 6	#44 ×	#45 ÷
#46 1	#47 2	#48 3
#49 +	#50 –	#51 0

#52 .	#53 (−) ⇧ Ans	#54 E
#55 EXE		

U-45 CALCULATORS · Calculator #39 · cut out and keep